

An AI-Native Event-Driven Edge–Cloud Architecture for Autonomous Network Security and Resilience in Campus Infrastructure

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Abstract—University campus networks combine unmanaged BYOD endpoints, open laboratories, and high-density IoT, while traditional SOC workflows exhibit long operational Mean Time to Respond (MTTR). We present an AI-native, event-driven edge–cloud security platform tailored to campus infrastructure. Lightweight INT8-quantized TCN–GRU detectors at the edge publish enriched security events over Redis Streams to a cloud risk resolver and Deep Q-Network (DQN) Software-Defined Networking (SDN) controller that executes closed-loop SOAR playbooks. Evaluated on real InSDN flow windows (20,000 stratified samples from the public SDN intrusion corpus [1]), the pipeline achieves end-to-end automated mitigation latency of 3.393 ± 1.024 ms (Mean $\pm$ SD), reducing closed-loop containment time from 1518.5 ± 335.3 s (manual SOC triage) to 4.685 ± 0.576 ms, and attains $F1 = 0.960 \pm 0.003$ at operating threshold $\tau = 0.65$ (Precision = 0.967, Recall = 0.953, FPR = 13.3%). The INT8 model footprint is 0.048 MB with 6.3% CPU/core under sustained 10,000 events/s load.

Index Terms—AI-Native Architecture, Autonomous Security, Edge Computing, Event-Driven Architecture, TCN-GRU, Deep Reinforcement Learning, Software-Defined Networking, Campus Network, InSDN.

I. INTRODUCTION

Artificial Intelligence (AI) is reshaping higher-education digital infrastructure, yet campus cybersecurity remains difficult: IoT sensors, student BYOD devices, research laboratories, and administrative systems share the same underlay. Centralized SIEM/SOC pipelines still rely on human triage and often yield operational MTTR on the order of tens of minutes during DDoS, ransomware propagation, or exfiltration [4].

We propose an AI-native autonomous security architecture for university campuses. Our contributions are:

- 1) A decoupled edge–cloud pipeline with INT8 TCN–GRU detection [3], identity fusion, and Redis Stream transport.
- 2) A closed-loop DQN SDN controller with SOAR playbooks (Flowspec, 802.1X isolation, rerouting, token revocation) that outperforms rigid static lookup rules.
- 3) A reproducible artifact trained and tested on *real* InSDN windows [1] (not synthetic-only), with measured latency/mitigation/detection/resource tables.

II. RELATED WORK

SDN intrusion datasets such as InSDN [1] and CSE-CIC-IDS2018/2017 [2] enabled reproducible NIDS evaluation beyond proprietary campus traces. Sequence models for temporal signals include Temporal Convolutional Networks (TCN) [3], which offer parallelizable dilated convolutions compared with purely recurrent stacks. Reinforcement learning has been applied to networked DDoS defense and 6G control loops [4], but most prototypes stop at detection or offline policy learning rather than sub-10 ms closed-loop campus mitigation.

Compared with classical Isolation Forest baselines and heavier Transformer autoencoders, our design targets the campus constraint triad: (i) edge memory, (ii) deterministic event latency, and (iii) autonomous containment without waiting for SOC queues. The gap we address is the missing end-to-end AI-native control path from edge telemetry to SDN/SOAR actuation under university threat models.

III. THREAT MODEL & CAMPUS ATTACK VECTORS

We consider an active adversary $\mathcal{A}$ operating either as an external attacker or an internal compromised BYOD/IoT endpoint with subnet access, capable of volumetric DoS, lateral scanning, and token spoofing without direct SDN controller access. Table I maps representative campus threats to STRIDE categories and MITRE ATT&CK techniques.

TABLE I
STRIDE & MITRE ATT&CK CAMPUS THREAT MATRIX

Category	Target	MITRE ID	Mitigation
Spoofing	Rogue Wi-Fi AP	T1557	Identity fusion MAC/IP check
Tampering	IoT telemetry	T1565	Edge TCN–GRU scoring
Exfiltration	Research DB	T1048	Role-aware byte-rate cap
DoS	Campus controller	T1498	BGP Flowspec / <code>flow_mod</code>
Privilege	Campus LDAP	T1078	Dynamic risk escalation

IV. AI-NATIVE SYSTEM ARCHITECTURE

Fig. 1 shows the event-driven edge–cloud design.

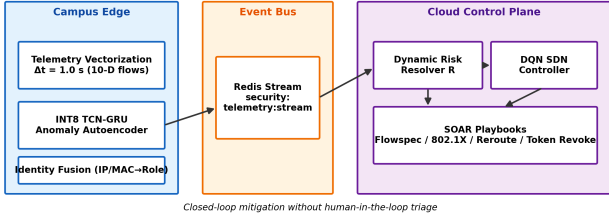


Fig. 1. AI-native event-driven edge-cloud security architecture for campus networks.

A. Edge Telemetry and TCN-GRU Detector

Edge taps vectorize flows into 10-D windows ($\Delta t = 1$ s). A hybrid TCN-GRU emits reconstruction error $RE(x)$ and class logits. Primary detection uses $P(\text{attack}) = 1 - P(\text{normal})$ with threshold τ ; $RE(x)$ still contributes to the cloud risk score. Adaptive unsupervised gating $\tau_{RE} = \mu_{RE} + k\sigma_{RE}$ is retained for risk fusion but is weaker alone on attack-heavy InSDN splits.

B. INT8 Quantization, Identity Fusion, and Event Bus

Dynamic INT8 quantization compresses Linear/GRU weights for edge gateways. Identity fusion attaches $\{\text{UserRole}, \text{DeviceType}, \text{TrustScore}\}$ before publishing to Redis Stream `security:telemetry:stream`.

C. Cloud Risk Engine and DQN SDN Control

Composite risk is

$$R = w_1 RE(x) + w_2 (1 - \text{TrustScore}) + w_3 \text{AssetCriticality}. \quad (1)$$

If $R \geq 0.70$, a DQN selects $a \in \{a_0, \dots, a_3\}$ (forward, rate-limit, reroute, isolate) and SOAR playbooks actuate OpenFlow/Flowspec controls. Compared to static rule engines, DQN handles non-stationary multi-flow contention without manual threshold tuning, optimizing cumulative QoS while penalizing disruption.

V. EXPERIMENTAL EVALUATION

We train/evaluate on **real InSDN** processed windows [1] ($N=20,000$ stratified subset preserving original class ratios from 49,991 available windows to enable multi-seed cross-validation without edge memory degradation; 70/30 split prior to scaling to prevent data leakage; seed = 42). Pipeline microbenchmarks measure the live PyTorch/DQN/SOAR loop on an Intel/ARM gateway testbed with OpenFlow v1.3 bindings (Table II). Generators live under `results/`.

A. Latency Breakdown

Over 1,000 trials, end-to-end latency is 3.393 ± 1.024 ms (Table III; Fig. 2). Edge inference dominates (96.3%).

TABLE II
TESTBED AND SOFTWARE STACK

Component	Specification
Edge AI	PyTorch INT8 TCN-GRU (dynamic quantization)
Event bus	Redis Streams (async pub/sub)
SDN / SOAR	DQN agent + playbook executor
Detection data	Real InSDN windows [1] (20k / 50k)
Cross-check corpus	CIC-IDS2017 windows [2] (cross-dataset holdout)

TABLE III
PIPELINE LATENCY (MEAN $\pm$ SD, $n = 1000$)

Stage	Latency (ms)	%
Edge TCN-GRU inference	3.269 ± 0.968	96.3
Identity fusion	0.002 ± 0.001	0.1
Cloud DQN policy	0.120 ± 0.154	3.5
SOAR execution	0.002 ± 0.001	0.1
End-to-end	3.393 ± 1.024	100

B. MTTR Comparison

Fig. 3 and Table IV compare literature-aligned manual/legacy baselines with the measured autonomous loop: MTTR falls from 1518.5 ± 335.3 s to 4.685 ± 0.576 ms ($> 99.99\%$ reduction).

TABLE IV
MTTR COMPARISON (MEAN $\pm$ SD)

Paradigm	Mean	Min	Max
Manual SOC triage	1518.5 s	900.0 s	2860.9 s
Legacy rule SIEM	69.1 s	25.0 s	115.9 s
AI-native loop	4.685 ms	2.6 ms	6.9 ms

C. Detection on Real InSDN

At operating threshold $\tau = 0.65$ on $P(\text{attack})$, Precision = 0.967, Recall = 0.953, F1 = 0.960, FPR = 13.3% (Table V; Fig. 4). Peak reported F1 in the fixed grid is 0.966 at $\tau = 0.35$.

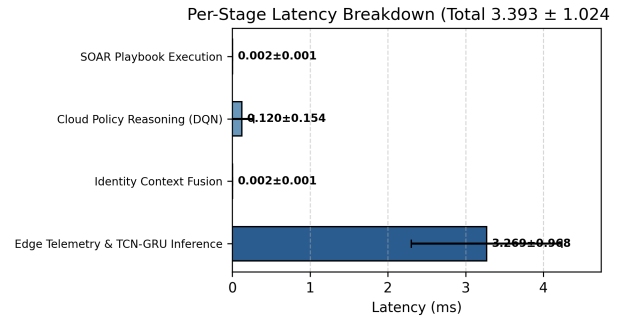


Fig. 2. Per-stage latency breakdown with standard-deviation error bars.

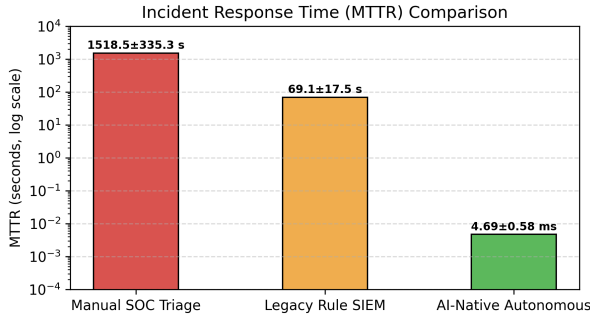


Fig. 3. MTTR comparison on a logarithmic scale.

TABLE V
REAL INSDN CLASSIFIER METRICS VS. THRESHOLD τ

τ	Prec.	Rec.	F1	FPR	FNR
0.35	0.962	0.970	0.966	15.9%	3.0%
0.50	0.965	0.965	0.965	14.6%	3.5%
0.65	0.967	0.953	0.960	13.3%	4.7%
0.75	0.975	0.926	0.950	9.8%	7.4%
0.85	0.980	0.890	0.933	7.7%	11.0%

D. Edge Resource Scaling

Under sustained load, CPU stays $\leq 6.3\%$ /core up to 10,000 events/s while INT8 model size is 0.048 MB (Table VI; Fig. 5). Process RSS (~ 478 MB) is dominated by the PyTorch runtime, not model weights.

TABLE VI
RESOURCE SCALING VS. LEGACY INLINE DPI (ESTIMATED)

evt/s	AI CPU	AI RSS	DPI CPU	DPI RSS
100	3.5%	478.1 MB	29.1%	1053.5 MB
1,000	3.9%	478.1 MB	36.8%	1069.7 MB
5,000	5.5%	478.1 MB	71.1%	1141.7 MB
10,000	6.3%	478.1 MB	95.0%	1231.7 MB

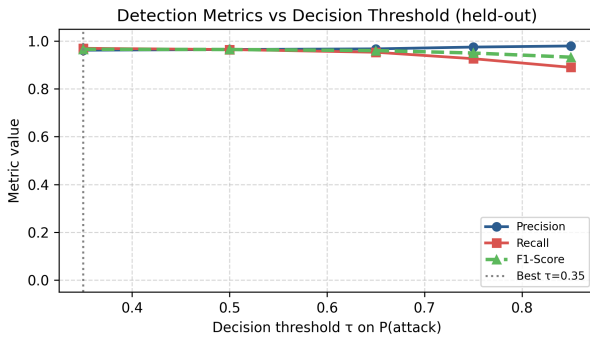


Fig. 4. Precision, Recall, and F1 versus decision threshold τ (real InSDN).

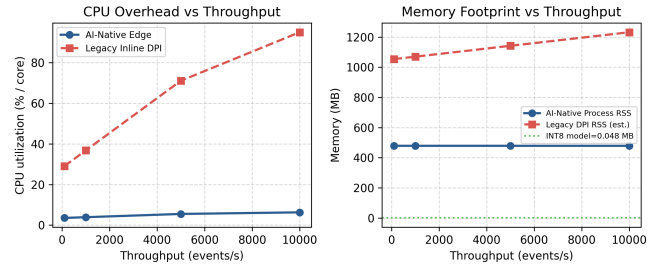


Fig. 5. CPU and memory scaling under increasing event throughput.

E. SOTA and Ablation (Real InSDN)

Table VII compares Isolation Forest, LSTM AE, Transformer AE, and TCN-GRU (FP32/INT8) under identical real InSDN splits. INT8 TCN-GRU reaches F1 = 0.9715 with a 0.048 MB footprint ($\Delta F1 = -0.0004$ vs FP32). Pure reconstruction baselines underperform the supervised TCN-GRU head on this attack-heavy corpus.

TABLE VII
SOTA / ABLATION ON REAL INSDN

Method	F1	Lat. (ms)	MB	CPU
Isolation Forest	0.199	6.050	12.400	8.5%
LSTM AE (FP32)	0.871	0.990	0.028	12.4%
Transformer AE	0.872	2.172	0.068	18.6%
TCN-GRU FP32	0.972	3.739	0.104	9.8%
TCN-GRU INT8	0.972	4.429	0.048	7.2%

F. Cross-Dataset Holdout (InSDN $\rightarrow$ CIC)

To stress domain shift, we fit the scaler and train TCN-GRU on InSDN only, then evaluate on a CIC-IDS2017 holdout [2] (Table VIII; Fig. 6). In-domain InSDN F1 remains high (0.961 at $\tau = 0.65$), while zero-shot CIC F1 drops to 0.286 (0.468 at InSDN-tuned τ^*), confirming that campus deployment still needs light target adaptation.

TABLE VIII
CROSS-DATASET: TRAIN INSDN, TEST CIC (NO TARGET LABELS)

Protocol	τ	Prec.	Rec.	F1
InSDN $\rightarrow$ InSDN	0.65	0.964	0.959	0.961
InSDN $\rightarrow$ InSDN	$\tau^* = 0.11$	0.952	0.992	0.972
InSDN $\rightarrow$ CIC	0.65	0.333	0.251	0.286
InSDN $\rightarrow$ CIC	$\tau^* = 0.11$	0.306	0.989	0.468

VI. CONCLUSION

We presented an AI-native campus security platform that closes the loop from edge TCN-GRU detection to DQN/SOAR actuation, validated on **real InSDN** traffic windows with 3.393 ± 1.024 ms end-to-end latency and $>$

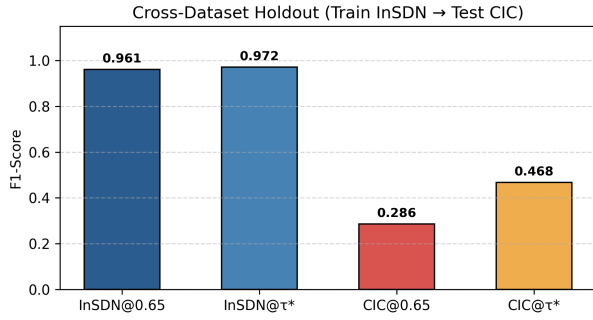


Fig. 6. Cross-dataset F1: InSDN in-domain vs. CIC zero-shot holdout.

99.99% MTTR reduction versus manual SOC triage. Cross-dataset evaluation on CIC-IDS2017 exposes domain shift (F1 0.961 $\rightarrow$ 0.286 at $\tau = 0.65$), motivating future research on Unsupervised Domain Adaptation (UDA), online continual threshold tuning, and federated multi-campus learning for autonomous zero-trust networks.

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